How Much Does Semantic Search Help? Twelve Retrieval Methods vs. BM25 on Four BEIR Collections, Checked Against Published Results

Spike T1 — CS410 Technology Review: LLMs and Semantic Search

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Abstract

We compare twelve retrieval methods against a fixed BM25 baseline on four zero-shot BEIR collections (SciFact, NFCorpus, FiQA-2018, TREC-COVID). The methods are learned sparse (SPLADE), dense bi-encoders (bge-small, Contriever), late interaction (ColBERTv2/PLAID), RRF fusion, cross-encoder reranking and two LLM methods (HyDE query expansion and RankGPT listwise reranking with gpt-oss-120b). Every method from a paper is first reproduced: BM25 (BEIR configuration), BM25+cross-encoder, SPLADE, ColBERTv2, bge-small and Contriever land within 0.014 nDCG@10 of their published numbers on every collection. HyDE does not reproduce with gpt-oss-120b and Contriever, and we show why. Against BM25, the semantic methods win clearly where queries are natural-language questions (FiQA, TREC-COVID: up to +0.18 and +0.27 nDCG@10) and modestly on SciFact and NFCorpus. RankGPT is the most effective system on three of four collections (0.878 on TREC-COVID, above the published gpt-4 result), and HyDE-bge is best on SciFact, but both cost 10–50 s per query. Hybrid RRF gives most of the gain for about 10 ms. Filling judgment holes with an LLM judge (Thomas et al.) widens the semantic methods' lead rather than closing it.

1 Why these collections

A first run of this spike on the CS410 MP collections (archived in `archive_mp/`) used short keyword queries and lexically pooled judgments, so it mostly tested semantic methods on BM25's home ground. This run uses four BEIR [18] test collections where query and relevant text tend to be worded differently (Table 1).

Table 1: Collections (BEIR test splits). All queries have at least one relevant document.

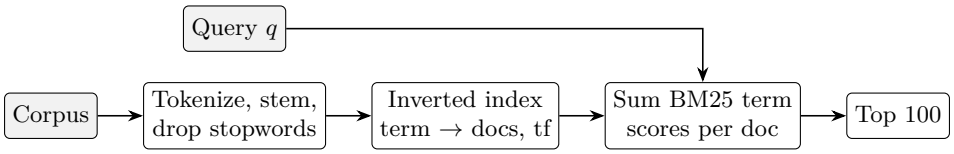
Collection	Docs	Queries	Qrels	Grades	What it tests
SciFact [21]	5,183	300	339	1	Scientific claims → supporting abstracts
NFCorpus [3]	3,633	323	12,334	1–2	Lay health queries → medical papers
FiQA-2018 [13]	57,638	648	1,706	1	Natural-language finance questions → forum answers
TREC-COVID [20]	171,332	50	66,336	0–2	COVID questions → CORD-19 abstracts; deep pools

2 How each method works

This section defines each method in a few sentences and shows its data flow. q is the query, d a document, and every method ends with a ranked list of document ids.

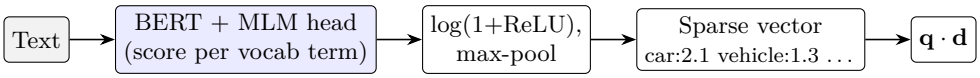
2.1 BM25 (lexical baseline) [9, 18]

BM25 scores a document by the query terms it contains. Each matching term contributes its inverse document frequency (rare terms count more), damped by term frequency (controlled by k_1) and normalized by document length (controlled by b): $\text{score}(q,d) = \sum_{t \in q} \text{idf}(t)\, \frac{tf_{t,d}(k_1+1)}{tf_{t,d}+k_1(1-b+b\,|d|/\overline{|d|})}$. Text is tokenized and stemmed first, so "running" matches "run", but "car" never matches "automobile". An inverted index makes this very fast. BM25-BEIR is the same formula with BEIR's parameters and with title and body scored separately and summed.



2.2 SPLADE (learned sparse retrieval) [5, 6]

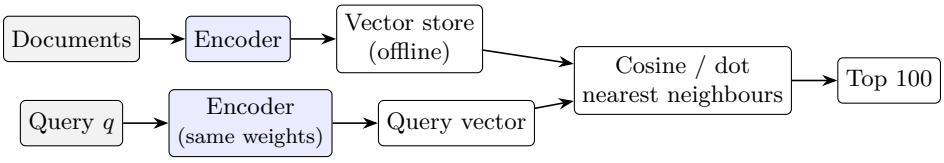
SPLADE keeps BM25's "bag of weighted terms" form but lets a BERT model choose the terms and weights. For every token position, the masked-language-model head predicts a score for each of the 30,522 vocabulary terms. These are passed through $\log(1 + \text{ReLU}(\cdot))$ and max-pooled over positions, giving one sparse vector per text. The vector also contains related terms that never appear in the text (*expansion*, e.g. "car" → "vehicle"). Relevance is the dot product of the query and document vectors, which an inverted index can compute.



same model for queries and documents

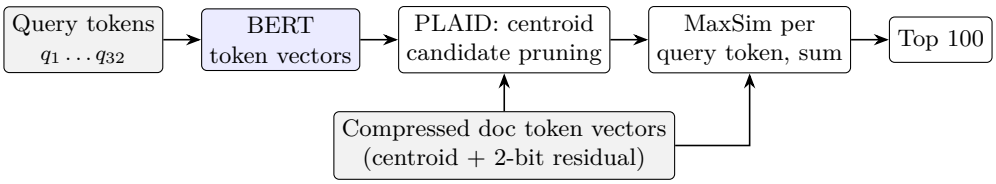
2.3 Dense bi-encoder (Dense = bge-small, Contriever) [8, 10, 23]

A transformer encodes the query and each document *separately* into a single dense vector (384 dimensions for bge-small, 768 for Contriever). Training places texts with the same meaning close together, so a query and a relevant document match even if they share no words. Relevance is cosine similarity (bge, normalized vectors) or a dot product (Contriever). Document vectors are computed once, offline; at query time only the query is encoded, followed by a nearest-neighbour search. bge-small is fine-tuned on large supervised relevance data; Contriever is trained without labels (contrastive learning on text crops), which explains its weaker results.



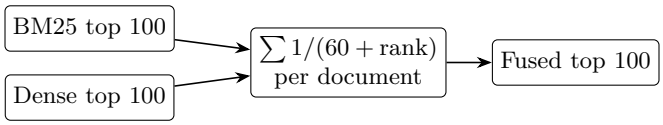
2.4 ColBERTv2 with PLAID (late interaction) [11, 14, 15]

ColBERT keeps one vector *per token* instead of one per text. A query is scored against a document by MaxSim: each query token finds its most similar document token, and these maxima are summed, score $= \sum_{i \in q} \max_{j \in d} \mathbf{q}_i \cdot \mathbf{d}_j$. This captures term-level matches with contextual meaning. Storing every token vector is expensive, so ColBERTv2 compresses each one to a cluster centroid plus a 2-bit residual. PLAID first uses the centroids to pick a small candidate set cheaply, then computes exact MaxSim only for those candidates.



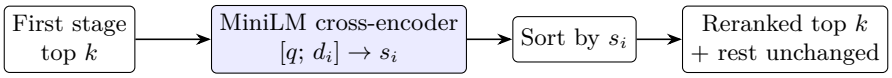
2.5 Hybrid: Reciprocal Rank Fusion [4]

RRF merges several ranked lists using only the rank positions, so scores on different scales (BM25 points vs. cosine similarities) never need calibrating. Each document receives $\sum_\ell 1/(60 + \text{rank}_\ell(d))$ over the lists that contain it. A document ranked highly by either list, or moderately by both, rises to the top. Here the lists are the BM25 and Dense top 100.



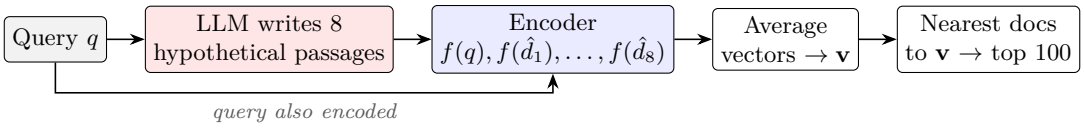
2.6 Cross-encoder reranking (Rerank, BM25+CE) [18, 22]

A cross-encoder reads the query and one document *together* in a single transformer pass ("[CLS] query [SEP] document"), so every query word can attend to every document word. It outputs one relevance score. This is more accurate than comparing separately computed vectors, but it must run once per (query, document) pair at query time, so it is applied only to a shortlist from a first-stage retriever: the Hybrid top 50 (Rerank) or the BM25-BEIR top 100 (BM25+CE).



2.7 HyDE*: Hypothetical Document Embeddings [7]

Idea. A dense retriever compares the query vector $f(q)$ with every document vector $f(d)$ and returns the nearest. But queries and documents are different kinds of text: a query is a few words ("Deleting Raptor reduces G-CSF levels."), a document is a full abstract with its own vocabulary. Their vectors live in different parts of the space, so the nearest documents to a *query* need not be the documents that *answer* it. HyDE turns retrieval into document–document similarity, which encoders are good at: an LLM *writes* N plausible answer passages $\hat{d}_1, \dots, \hat{d}_N$ ("Please write a scientific paper passage to support/refute the claim"), and the search looks for real documents that are close to those fake ones. The passages may contain invented facts; that does not matter, because they are never shown to the user, only encoded. The LLM never sees the corpus.

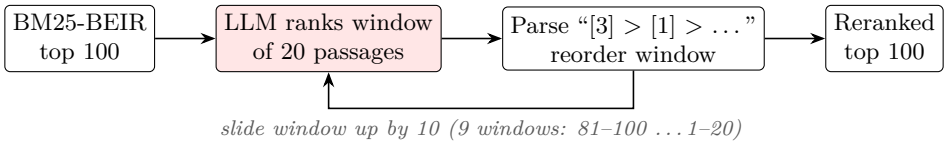


The search vector is the mean of $f(q)$ and the eight passage vectors, and the search itself is the same nearest-neighbour search as Dense.

** HyDE is the method we plan to use for our project, CCDB Search (semantic search over the CFPB Consumer Complaint Database). How the averaged vector finds the right documents, our measurements of it, and what they mean for CCDB Search are in Section 6.*

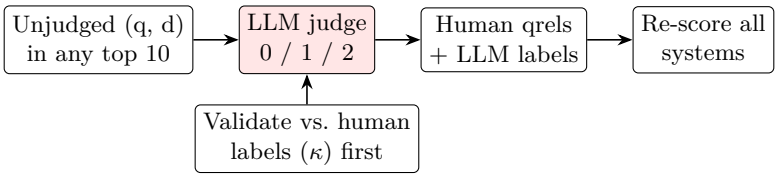
2.8 RankGPT: listwise LLM reranking [17]

RankGPT gives an LLM the query and 20 numbered passages and asks it to output a ranking such as "[3] > [1] > [7] > ...". Because 100 passages do not fit well in one prompt, it uses a sliding window: rerank positions 81–100, slide up by 10 to rerank 71–90, and so on to the top. Strong documents can thus "bubble up" from the bottom of the list. That makes 9 LLM calls per query, which must run in sequence. Unparseable or missing ids are appended in their original order.



2.9 LLM relevance judge (evaluation aid, not a retriever)

Human relevance labels exist only for documents that were in the original judgment pool, and unjudged documents count as non-relevant. Following Thomas et al. [19], an LLM (Qwen3.5) is shown the query and one document and asked for a 0–2 relevance grade, after considering intent match (M) and trustworthiness (T). We first measure its agreement with human labels, then use it only to label the unjudged documents in each system's top 10. Human labels are never overwritten.



3 Systems

There is one script per method; all see the same corpus (title + body) and the test queries, and each returns a top-100 list. Nothing was tuned on the test queries. Section 2 explains how each method works; Table 2 lists the exact configurations; "repro." marks a system configured to match a published result.

Table 2: Systems. LLM calls go through the Databricks Unity Gateway; every call is cached on disk. * The method planned for CCDB Search over the CFPB Consumer Complaint Database (Section 6).

System	Source	Configuration
BM25 (baseline)	BEIR [18], [9]	`bm25s`, Lucene variant, k_1=1.2, b=0.75, stopwords, Snowball stemmer
BM25-BEIR (repro.)	BEIR [18]	Anserini setting: k_1=0.9, b=0.4, Porter stemmer, title and body indexed as separate fields with scores summed
SPLADE (repro.)	[5, 6]	`naver/splade_v2_distil`, 256 tokens, sparse dot product
Dense (repro.)	DPR [10]; [2, 23]	`bge-small-en-v1.5`, CLS pooling, normalized, query instruction prefix, exact cosine search
Contriever (repro.)	[8], HyDE T2 [7]	`facebook/contriever` (unsupervised), mean pooling, unnormalized dot product
ColBERTv2 (repro.)	[11, 14, 15]	`colbertv2.0` via PyLate, PLAID index, 2-bit residuals, doc 300 / query 32 tokens
Hybrid	Cormack et al. [4]	RRF of BM25 and Dense, $\sum 1/(60 + r)$ with r from 1 as in the paper
Rerank	BEIR [18], [22]	`ms-marco-MiniLM-L-6-v2` rescores the Hybrid top 50
BM25+CE (repro.)	BEIR T2 [18]	Same cross-encoder on the BM25-BEIR top 100
HyDE-bge*	HyDE [7]	gpt-oss-120b writes 8 passages per query (temp. 0.7, 512 tokens, the paper's prompts); query vector = mean of query and passage embeddings; bge encoder
HyDE-Contriever (repro.)	HyDE [7]	As above with the paper's Contriever encoder
HyDE-Contriever-100w	ours	As above, but passages are stripped of markdown and cut to 100 words before encoding (Section 4.1)
RankGPT (repro.)	Sun et al. [17]	Official prompts, passages cut to 300 words, sliding window 20 / step 10 over the BM25-BEIR top 100; gpt-oss-120b, temperature 0

Implementation notes from checking each paper:

- BEIR's BM25 baseline is Anserini/Lucene with k_1=0.9, b=0.4 and multi-field indexing. Our default BM25 keeps the common k_1=1.2, b=0.75; it scores slightly higher on SciFact and FiQA and lower on TREC-COVID, so we report both.
- PyTerrier's `RRFusion` [1, 12] uses 0-based ranks, i.e. $1/(k + r - 1)$ with 1-based r; we follow the original paper.
- DPR used an unnormalized dot product, FAISS HNSW search and a linear BM25 hybrid (λ=1.1). We use cosine scoring as specified by the bge model card, exact search, and RRF.
- SPLADE scores 0.703 on SciFact at 512 tokens; the paper's 256-token setting gives 0.693, matching the published number exactly.
- The LLMs differ from the papers': gpt-oss-120b (reasoning effort "low") replaces text-davinci-003 (HyDE) and gpt-3.5/gpt-4 (RankGPT). The judge is Qwen3.5-122B-A10B with thinking off, deliberately a different model from the one doing the reranking.

Evaluation. `ir-measures` (pytrec_eval) on the graded test qrels: nDCG@10 (primary), Recall@100 (secondary), P@10 and AP@100; unjudged documents count as non-relevant, as in BEIR. R_{cap}@100 divides by $\min(|\text{rel}|, 100)$, as in BEIR Table 9. Each system is compared with BM25 on per-query nDCG@10 with a paired t-test. BEIR runs no significance tests; the test is our addition, following Smucker et al. [16]. Latency is the mean wall-clock time per query on an Apple M5 Max (PyTorch MPS), model load excluded. Seeds are fixed at 0.

4 Reproductions against published results

Table 5 puts each reproduction next to the number its paper reports. Six of seven methods reproduce: the largest gap is 0.013 (BM25-BEIR on TREC-COVID), and SPLADE, bge-small and Contriever agree to within 0.003 on every collection. BM25-BEIR also matches BEIR's capped Recall@100 (TREC-COVID 0.499 vs. 0.498), and the cross-encoder setup matches BEIR's BM25+CE rows to within 0.005. These agreements matter because they show the pipeline, data and metric code are right, so the differences in Section 5 come from the methods.

4.1 HyDE does not reproduce with gpt-oss

HyDE-Contriever reaches only 0.556 / 0.063 / 0.434 (SciFact / FiQA / TREC-COVID) against the published 0.691 / 0.273 / 0.593, which is *below* Contriever alone. Contriever itself reproduces exactly, so the cause is in the generated passages. gpt-oss-120b writes long answers (median 507 words on FiQA, many with markdown headings and tables), far longer than the paper's passages. Contriever's mean-pooled vectors are not normalized, so these long, formatted passages dominate the averaged query vector. Stripping markdown and cutting each passage to 100 words (HyDE-Contriever-100w) recovers SciFact (0.684) and FiQA (0.285, above the published 0.273) but hurts TREC-COVID (0.351), so it is not a general fix, and we report it as our variant, not a reproduction. With the normalized bge encoder the same passages work well: HyDE-bge improves on Dense on SciFact (+0.056), NFCorpus (+0.021) and FiQA (+0.007) and ties on TREC-COVID (−0.002). HyDE is therefore sensitive to the generator's output style in a way the paper, using one generator, did not have to address.

4.2 RankGPT with gpt-oss beats the published gpt-3.5 results

Reranking the same BM25 top 100, RankGPT with gpt-oss-120b scores 0.765 / 0.383 / 0.878 (SciFact / NFCorpus / TREC-COVID), against 0.704 / 0.356 / 0.767 for gpt-3.5-turbo and 0.750 / 0.385 / 0.855 for gpt-4 in the paper. The gpt-4 rows reranked gpt-3.5's top 30, not BM25's top 100, so that comparison is not like-for-like. Our BM25-BEIR first stage is also stronger than the paper's Pyserini BM25 on TREC-COVID (0.669 vs. 0.595), which accounts for part of the gap. Output problems were rare: across 11,022 windows, 12 duplicate ids, 994 missing ids (appended in original order by the official `receive_permutation`) and 4 empty responses.

5 Results against BM25

Table 3: nDCG@10 per system (rows) and collection (columns). Δ = difference from BM25; * paired t-test $p < 0.05$, ** $p < 0.001$. Best per column in bold.

		scifact		nfcorpus		fiqa		trec-covid	
System	Family	nDCG	Δ	nDCG	Δ	nDCG	Δ	nDCG	Δ
BM25	Lexical	0.679	–	0.318	–	0.253	–	0.609	–
BM25-BEIR	Lexical	0.656	-0.023*	0.320	+0.002	0.236	-0.017*	0.669	+0.060*
SPLADE	Learned sparse	0.693	+0.013	0.334	+0.016	0.336	+0.083**	0.710	+0.101*
Dense	Dense	0.713	+0.034*	0.343	+0.025*	0.403	+0.150**	0.757	+0.147**
Contriever	Dense	0.649	-0.030	0.317	-0.001	0.245	-0.008	0.274	-0.335**
ColBERTv2	Late interaction	0.691	+0.012	0.339	+0.021*	0.355	+0.102**	0.731	+0.121*
HyDE-bge	LLM	**0.769**	+0.090**	0.364	+0.046**	0.410	+0.157**	0.755	+0.146**
HyDE-Contriever	LLM	0.556	-0.123**	0.191	-0.128**	0.063	-0.190**	0.434	-0.175**
HyDE-Contriever-100w	LLM	0.684	+0.005	0.289	-0.030*	0.285	+0.032*	0.351	-0.259**
Hybrid	Hybrid	0.724	+0.045**	0.361	+0.042**	0.366	+0.113**	0.778	+0.169**
Rerank	Reranker	0.697	+0.017	0.357	+0.038**	0.371	+0.118**	0.779	+0.170**
BM25+CE	Reranker	0.689	+0.010	0.350	+0.032**	0.351	+0.098**	0.762	+0.152**
RankGPT	LLM	0.765	+0.086**	**0.383**	+0.064**	**0.429**	+0.176**	**0.878**	+0.268**

Table 4: Secondary metrics. R_{cap}@100 divides by $\min(|rel|, 100)$ as in BEIR Table 9; J@10 is a diagnostic (share of top-10 docs with any human judgment).

	scifact					nfcorpus					fiqa					trec-covid				
System	R@100	R_{cap}	P@10	AP	J@10	R@100	R_{cap}	P@10	AP	J@10	R@100	R_{cap}	P@10	AP	J@10	R@100	R_{cap}	P@10	AP	J@10
BM25	0.913	0.913	0.089	0.638	0.089	0.247	0.260	0.229	0.143	0.229	0.553	0.553	0.070	0.205	0.070	0.110	0.472	0.642	0.074	0.876
BM25-BEIR	0.905	0.905	0.088	0.612	0.088	0.250	0.263	0.231	0.143	0.231	0.542	0.542	0.064	0.193	0.064	0.114	0.499	0.724	0.081	0.906
SPLADE	0.920	0.920	0.090	0.661	0.090	0.277	0.293	0.240	0.157	0.240	0.621	0.621	0.090	0.280	0.090	0.123	0.550	0.764	0.088	0.940
Dense	0.942	0.942	0.094	0.674	0.094	0.311	0.326	0.257	0.156	0.257	0.696	0.696	0.111	0.346	0.111	0.133	0.561	0.794	0.099	0.878
Contriever	0.926	0.926	0.089	0.606	0.089	0.294	0.308	0.231	0.148	0.231	0.562	0.562	0.069	0.199	0.069	0.037	0.172	0.270	0.016	0.346
ColBERTv2	0.922	0.922	0.090	0.657	0.090	0.288	0.303	0.238	0.159	0.238	0.638	0.638	0.094	0.299	0.094	0.128	0.555	0.794	0.094	0.952
HyDE-bge	0.983	0.983	0.102	0.727	0.102	0.333	0.347	0.275	0.166	0.275	0.727	0.727	0.112	0.352	0.112	0.139	0.584	0.796	0.109	0.876
HyDE-Contriever	0.886	0.886	0.079	0.508	0.079	0.212	0.220	0.145	0.072	0.145	0.187	0.187	0.016	0.050	0.016	0.064	0.296	0.466	0.034	0.546
HyDE-Contriever-100w	0.957	0.957	0.093	0.634	0.093	0.297	0.308	0.213	0.124	0.213	0.649	0.649	0.083	0.231	0.083	0.051	0.241	0.374	0.025	0.442
Hybrid	0.968	0.968	0.096	0.681	0.096	0.312	0.328	0.263	0.174	0.263	0.693	0.693	0.104	0.305	0.104	0.133	0.554	0.808	0.100	0.934
Rerank	0.968	0.968	0.093	0.656	0.093	0.312	0.328	0.251	0.173	0.251	0.693	0.693	0.104	0.310	0.104	0.133	0.554	0.858	0.102	0.974
BM25+CE	0.905	0.905	0.091	0.649	0.091	0.250	0.263	0.244	0.161	0.244	0.542	0.542	0.094	0.287	0.094	0.114	0.499	0.824	0.089	0.962
RankGPT	0.905	0.905	0.098	0.731	0.098	0.250	0.263	0.273	0.172	0.273	0.542	0.542	0.108	0.360	0.108	0.114	0.499	0.924	0.095	0.976

Table 5: Our runs vs. published results (nDCG@10 unless noted). "Ours" is the matching system from this study; Δ = ours − published. Rows without a match are reference points only.

		scifact		nfcorpus		fiqa		trec-covid	
Published (source)	Ours	pub.	Δ	pub.	Δ	pub.	Δ	pub.	Δ
BM25, Anserini k1=0.9 b=0.4 (BEIR T2) [18]	BM25-BEIR	0.665	-0.009	0.325	-0.005	0.236	+0.000	0.656	+0.013
BM25 top-100 + MiniLM-L6 CE (BEIR T2) [18]	BM25+CE	0.688	+0.001	0.350	+0.000	0.347	+0.004	0.757	+0.005
bge-small-en-v1.5 (model card, MTEB) [2]	Dense	0.713	-0.000	0.343	+0.000	0.403	+0.000	0.759	-0.002
DistilSPLADE-max (SPLADE v2 T2) [5]	SPLADE	0.693	-0.000	0.334	+0.000	0.336	-0.000	0.710	+0.000
ColBERTv2 (ColBERTv2 T5a) [15]	ColBERTv2	0.693	-0.002	0.338	+0.001	0.356	-0.001	0.738	-0.007
Contriever, unsupervised (HyDE T2) [7]	Contriever	0.649	+0.000	–		0.245	-0.000	0.273	+0.001
HyDE, text-davinci-003 + Contriever (HyDE T2) [7]	HyDE-Contriever	0.691	-0.135	–		0.273	-0.210	0.593	-0.159
RankGPT, gpt-3.5-turbo on BM25 top-100 (RankGPT T1) [17]	RankGPT	0.704	+0.060	0.356	+0.027	–		0.767	+0.111
BM25, Pyserini (RankGPT T1) [17]	–	0.679		0.307		–		0.595	
RankGPT, gpt-4 on gpt-3.5 top-30 (RankGPT T1) [17]	–	0.750		0.385		–		0.855	
monoT5 220M on BM25 top-100 (RankGPT T1) [17]	–	0.734		0.374		–		0.783	
DPR, NQ-trained (BEIR T2) [18]	–	0.318		0.189		0.112		0.332	
ColBERT v1 (BEIR T2) [18]	–	0.671		0.305		0.317		0.677	
Recall@100									
BM25 (BEIR T9) [18]	BM25-BEIR	0.908	-0.003	0.250	+0.013	0.539	+0.003	0.498	+0.001
bge-small-en-v1.5 (model card, MTEB; uncapped) [2]	Dense	0.942	-0.000	0.311	-0.000	0.696	+0.001	0.133	+0.000

Table 6: nDCG@10 with LLM labels filling the judgment holes (Thomas et al. [19] prompt, Qwen3.5-122B judge). Δ vs. BM25 under the same augmented qrels; * $p < 0.05$, ** $p < 0.001$.

	scifact			nfcorpus			fiqa			trec-covid		
System	orig.	aug.	Δ	orig.	aug.	Δ	orig.	aug.	Δ	orig.	aug.	Δ
BM25	0.679	0.705	–	0.318	0.508	–	0.253	0.506	–	0.609	0.624	–
BM25-BEIR	0.656	0.661	-0.045**	0.320	0.491	-0.017*	0.236	0.537	+0.031**	0.669	0.675	+0.051*
SPLADE	0.693	0.687	-0.018	0.334	0.543	+0.035**	0.336	0.603	+0.096**	0.710	0.719	+0.095*
Dense	0.713	0.712	+0.007	0.343	0.537	+0.029*	0.403	0.636	+0.130**	0.757	0.793	+0.169**
Contriever	0.649	0.671	-0.034*	0.317	0.500	-0.008	0.245	0.510	+0.004	0.274	0.329	-0.295**
ColBERTv2	0.691	0.687	-0.018	0.339	0.538	+0.030*	0.355	0.622	+0.116**	0.731	0.741	+0.117*
HyDE-bge	0.769	0.793	+0.087**	0.364	0.547	+0.039*	0.410	0.751	+0.245**	0.755	0.829	+0.205**
HyDE-Contriever	0.556	0.592	-0.113**	0.191	0.315	-0.193**	0.063	0.199	-0.307**	0.434	0.636	+0.012
HyDE-Contriever-100w	0.684	0.721	+0.016	0.289	0.440	-0.068**	0.285	0.652	+0.146**	0.351	0.504	-0.120*
Hybrid	0.724	0.743	+0.038**	0.361	0.573	+0.065**	0.366	0.639	+0.132**	0.778	0.793	+0.169**
Rerank	0.697	0.711	+0.006	0.357	0.566	+0.058**	0.371	0.651	+0.145**	0.779	0.790	+0.166**
BM25+CE	0.689	0.696	-0.010	0.350	0.552	+0.044**	0.351	0.622	+0.116**	0.762	0.766	+0.142**
RankGPT	0.765	0.818	+0.113**	0.383	0.619	+0.111**	0.429	0.799	+0.293**	0.878	0.892	+0.268**

Table 7: Mean end-to-end latency per query (ms; model load excluded; LLM methods include the gateway call time) and one-time index/encode cost (s).

	scifact		nfcorpus		fiqa		trec-covid	
System	ms/q	setup s	ms/q	setup s	ms/q	setup s	ms/q	setup s
BM25	0.2	0	0.1	0	0.5	2	1.3	7
BM25-BEIR	0.2	0	0.1	0	0.4	2	1.7	8
SPLADE	10.6	20	8.3	14	34.3	168	83.2	588
Dense	5.4	14	5.4	10	6.9	92	9.1	380
Contriever	6.6	40	6.2	31	8.6	276	11.6	1,037
ColBERTv2	255.0	113	162.2	48	259.8	1,006	383.6	5,397
HyDE-bge	12,338.5	14	19,194.9	10	13,451.1	93	9,555.9	354
HyDE-Contriever	12,383.6	40	19,239.6	31	13,495.7	276	9,601.6	1,037
HyDE-Contriever-100w	12,341.9	40	19,197.5	31	13,450.0	276	9,560.0	1,037
Hybrid	5.6	14	5.5	10	7.4	94	10.4	387
Rerank	86.8	14	91.3	10	93.6	94	94.2	387
BM25+CE	203.2	0	188.5	0	151.4	2	252.3	8
RankGPT	43,859.7	0	35,855.8	0	49,051.0	2	25,639.8	8

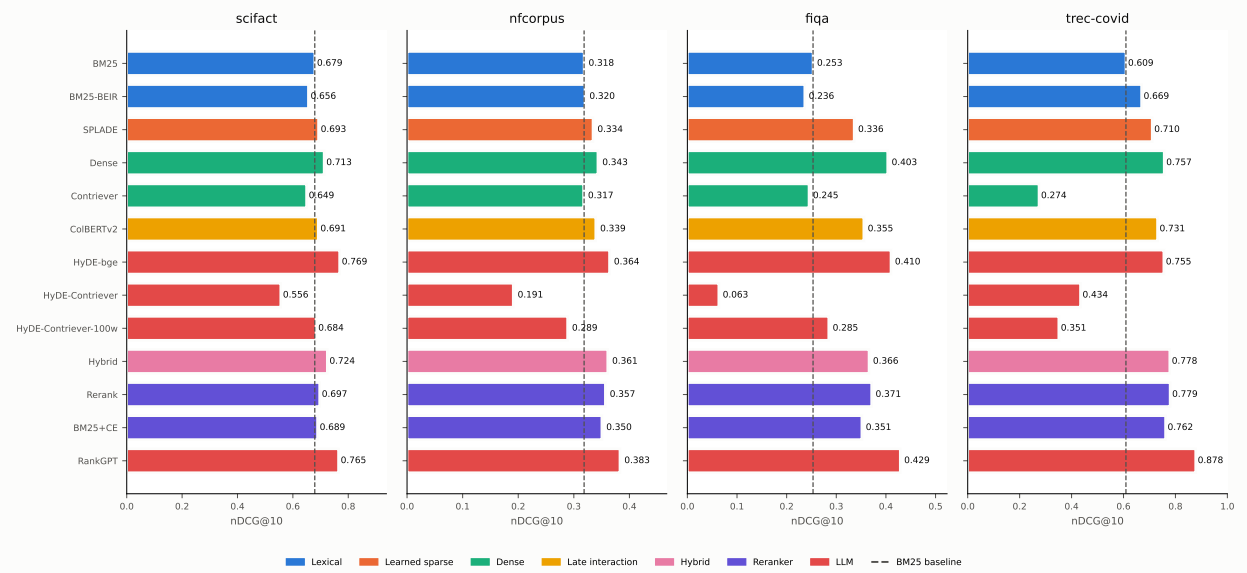


Figure 1: nDCG@10 per system and collection; dashed line = BM25 baseline.

Natural-language questions gain most. On FiQA every system built on a supervised encoder beats BM25 with $p < 0.001$: Dense +0.150 (313 wins, 86 losses), HyDE-bge +0.157, and RankGPT +0.176. On TREC-COVID, Dense, Hybrid, Rerank, HyDE-bge and RankGPT gain +0.15 to +0.27; RankGPT wins 43 of 50 queries. On SciFact (declarative claims sharing technical terms with abstracts) and NFCorpus (short topic titles), the gains are +0.01 to +0.09, and several systems are not significantly better than BM25.

Supervised training transfer matters more than architecture. Unsupervised Contriever never beats BM25 and collapses on TREC-COVID (0.274), as in the HyDE paper. SPLADE and ColBERTv2, both trained on MS MARCO, beat BM25 on FiQA and TREC-COVID but trail the

much smaller bge-small, whose training mixture is broader. The bge release does not document its English training data at the dataset level, so some overlap with these collections cannot be ruled out (Section 8).

Hybrid is the cheap, reliable choice. RRF of BM25 and Dense is significantly better than BM25 with $p < 0.001$ on all four collections and is within 0.10 of the best system everywhere, at about 10 ms per query. Its one weakness is FiQA, where equal-weight fusion with the weak BM25 list lowers nDCG@10 from Dense's 0.403 to 0.366.

Rerankers: the small cross-encoder helps a little, the LLM a lot. The MS MARCO MiniLM cross-encoder adds +0.01 to +0.15 over BM25 but at most +0.005 over Hybrid. RankGPT on the same BM25 list as BM25+CE is far better on every collection (+0.076 on SciFact, +0.078 on FiQA, +0.116 on TREC-COVID).

Recall@100. Semantic first stages miss fewer relevant documents (Table 4), e.g. FiQA 0.553 → 0.727 (HyDE-bge) and NFCorpus 0.247 → 0.333. The rerankers cannot raise Recall@100 above that of their first stage, which matters for retrieval-augmented generation, where the LLM only sees what the first stage retrieved.

5.1 Judgment holes: does pooling bias favour BM25?

Semantic systems retrieve many documents that were never judged; scoring them as non-relevant could understate those systems. Following Thomas et al. [19], we labelled every unjudged document in any system's top 10 with an LLM judge using the paper's prompt: 63,029 pairs in total, 18 of which failed and stay unjudged.

Validation first. On 200 stratified human-judged TREC-COVID pairs, the judge agrees with humans at Cohen's $\kappa = 0.53$ (binary and 3-class linear-weighted) and 62.5% exact agreement, which is moderate agreement. The other qrels contain only relevant documents, so κ is undefined there; the judge labels 80% (SciFact) and 91% (FiQA) of known-relevant documents as relevant, but only 27% on NFCorpus, whose relevance labels come from citation links rather than direct reading. The NFCorpus augmented numbers should therefore not be relied on.

Results. Table 6 rescored all systems with human labels plus LLM labels for the holes. All scores rise, because the holes contain relevant documents, and the semantic and LLM systems gain the most: on FiQA, BM25 rises 0.253 → 0.506 while HyDE-bge rises 0.410 → 0.751 and RankGPT 0.429 → 0.799; on TREC-COVID, Dense's gain over BM25 grows from +0.147 to +0.169. So pooling bias does affect these comparisons, and if anything it understates the semantic methods. One caveat applies to the LLM methods: an LLM judge may favour documents another LLM picked, even a different model.

6 HyDE* for CCDB Search

HyDE is the retrieval method we plan to use for our project, **CCDB Search**: semantic search over the CFPB Consumer Complaint Database (CCDB), the U.S. Consumer Financial Protection Bureau's public collection of consumer complaints about financial products and services such as credit reporting, debt collection, mortgages, credit cards and bank accounts. Many complaints include a free-text narrative written by the consumer (published only with the consumer's consent and with personal details scrubbed). These narratives are what CCDB Search will retrieve. This

section explains in detail how it finds documents (Section 6.1), shows what it does on the four test collections (Section 6.2), and turns the results into design choices for CCDB Search (Section 6.3).

We chose it for three reasons from this study. (i) It was the strongest single first-stage retriever on three of the four collections: HyDE-bge has the highest nDCG@10 of any non-reranking system on SciFact, NFCorpus and FiQA, and beats BM25 on all four collections. (ii) It needs no training and no labelled data, which matters for a new collection with no relevance judgments. (iii) It reuses an ordinary dense index: the LLM only rewrites the query, so the corpus is embedded once and never sent to the LLM.

6.1 How the averaged vector finds the right documents

The averaged vector. The search vector is the mean of the query embedding and the N passage embeddings (the paper's Eq. 8; we use $N = 8$, as the official code does):

$$\mathbf{v} \;=\; \frac{1}{N+1}\Big(f(q) + \sum_{k=1}^{N} f(\hat{d}_k)\Big).$$

Figure 2 draws this in a two-dimensional cartoon of the embedding space. Real documents form a region (gray); the short query lands outside it. Each hypothetical passage is written like a real document, so its vector lands *inside* the document region, near the documents that discuss the same topic. The passages differ in their invented details, so their vectors scatter, but they share the topic, so they scatter around the same place. Averaging cancels the sample-specific noise and keeps the shared direction; adding $f(q)$ keeps the vector anchored to what was actually asked. Search then returns the documents with the smallest angle to $\mathbf{v}$.

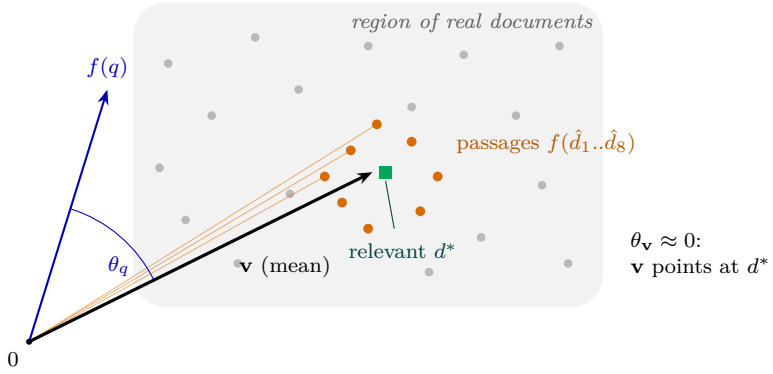
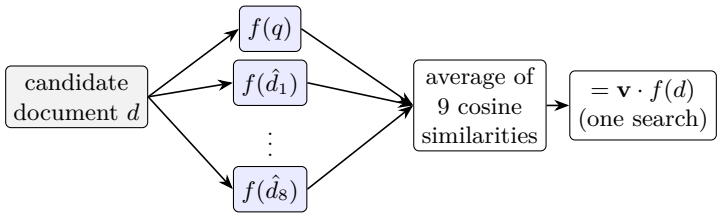


Figure 2: Cartoon of HyDE in embedding space. The query vector $f(q)$ points away from the document region; the eight hypothetical passages (orange) point into it, around the relevant document. Their mean $\mathbf{v}$ (black) has a much smaller angle to the relevant document ($\theta_{\mathbf{v}} \ll \theta_q$), so nearest-neighbour search on $\mathbf{v}$ ranks it near the top. Figure 3 shows the same picture for a real query.

Why averaging works: the score is an average of similarities. The inner product is linear, so scoring a document against the mean vector is the same as averaging its similarity to each of the nine vectors:

$$\text{score}(d) \;=\; \mathbf{v} \cdot f(d) \;=\; \frac{1}{N+1}\Big(f(q)\cdot f(d) \;+\; \sum_{k=1}^{N} f(\hat{d}_k)\cdot f(d)\Big).$$

So HyDE is a vote: a document ranks high only if it is close to the query *and* to most of the passages. A document that matches one passage's invented detail gets one vote; a document on the topic all passages share gets all of them. That is why hallucinated facts do little harm, and why the method still needs only one nearest-neighbour search (the mean is computed once, so the cost is the same as Dense search).



Why vector length matters. The mean treats vectors as arrows, so a longer arrow pulls harder. bge normalizes every vector to length 1, so the query and each passage get an equal 1/9 say. Contriever does not normalize; its vectors grow with text length. The cartoon below shows the effect we measured (Section 6.2): long LLM passages produce long vectors that swamp $f(q)$, and the mean drifts toward whatever the passages over-emphasize.



6.2 What the averaged vector does on real data

To see this mechanism in the real system, we re-encoded the cached gpt-oss passages and measured, for every query with a judged-relevant document, how the query vector $f(q)$, the individual passage vectors $f(\hat{d}_k)$ and the HyDE vector $\mathbf{v}$ relate to the relevant documents. The script is `retrieval_eval.evaluation.hyde_geometry`; it makes no new LLM calls. Table 8 summarizes it.

Worked example. SciFact claim 324 is "Deleting Raptor reduces G-CSF levels." Its relevant abstract never uses the word *Raptor*; it is about "mTOR signalling" regulating "G-CSF" in mammary tumours. Searching with $f(q)$ ranks it 113th, and none of the query's top 10 is relevant. The first gpt-oss passage begins "Raptor, a core component of the mechanistic target of rapamycin complex 1 (mTORC1)… granulocyte-colony stimulating factor (G-CSF)…": it is partly invented (it describes a mouse study that does not exist), but it supplies the missing link *Raptor* $\rightarrow$ *mTOR* in the vocabulary of an abstract. All eight passages do this in slightly different words. Their mean moves the search vector into the right neighbourhood: cosine to the relevant abstract rises from 0.646 to 0.836 and it is ranked **1st**. Figure 3 projects this query's vectors onto their first two principal components: the query's top-10 neighbours (blue rings) sit in the wrong area, while the eight passages cluster next to the relevant document and the top 10 around their mean (black rings) includes it.

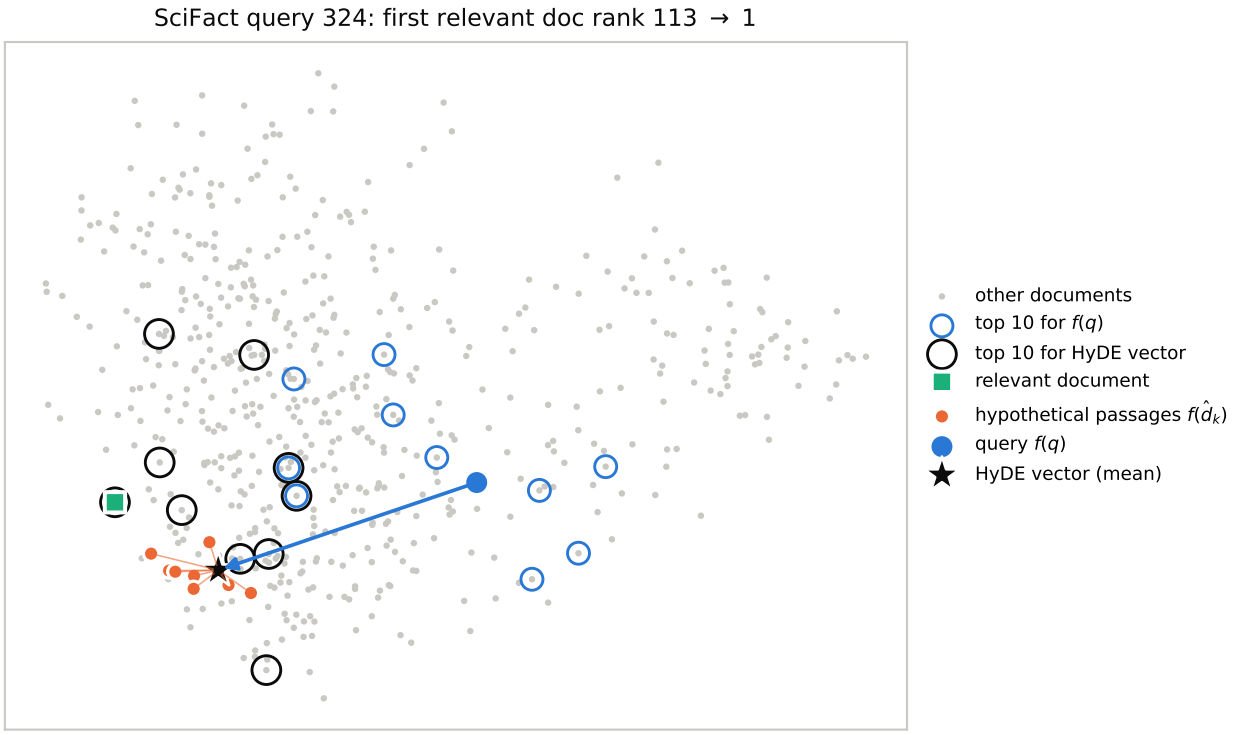


Figure 3: Real bge vectors for SciFact claim 324, projected to 2-D with PCA (fit on the plotted points: 300 nearest documents to $\mathbf{v}$, the query's top 10 and 300 random documents). The blue arrow is the move from $f(q)$ to the HyDE mean $\mathbf{v}$; thin orange lines join each passage to the mean. A 2-D projection distorts distances, so the rings (true top 10 in 384 dimensions) are the ground truth. Chosen as the query with the largest rank gain, so it is an illustration, not a typical case.

What the numbers show (bge).

- **The passages land in document space.** Each passage is closer to the relevant documents than the query is (mean cosine 0.869 vs. 0.773 on SciFact, 0.718 vs. 0.606 on NFCorpus), and the mean $\mathbf{v}$ is closer still (0.879, 0.730): averaging removes per-sample noise.
- **The eight passages are nearly the same vector.** Their pairwise cosine is 0.96 on every dataset, so the 8 samples mostly add robustness. The diversity the paper hopes for is small with gpt-oss at temperature 0.7.
- **Closer to everything, not only to the right thing.** $\mathbf{v}$ is also closer to random documents (mean cosine 0.56 → 0.65 on SciFact), because document-like vectors share a common direction. What matters for search is the *order*, so Figure 4 compares ranks. The first relevant document moves up for 29 % of SciFact queries and down for 12 %; on NFCorpus and FiQA the shifts are larger in both directions (31/25 % and 34/31 %); on TREC-COVID the first relevant document is usually already at rank 1. This matches the nDCG@10 gains: large on SciFact (+0.056), small on FiQA (+0.007).

Why Contriever fails. The same passages hurt Contriever, and the norms explain it. Contriever's query vectors have length ≈ 1.35, the passages ≈ 2.0 and corpus documents ≈ 1.75. In the unnormalized mean the eight long passages dominate: $\mathbf{v}$ keeps only a cosine of 0.51–0.62 with $f(q)$,

		mean cos to relevant docs				median rank of 1st rel.		% queries	passage
Encoder	Dataset	$f(q)$	$f(\hat{d})$	HyDE	$\cos(f(q), f(\hat{d}))$	$f(q)$	HyDE	rank better	words
bge	scifact	0.773	0.869	0.879	0.864	1	1	29	407
bge	nfcorpus	0.606	0.718	0.730	0.810	2	2	31	417
bge	fiqa	0.731	0.798	0.814	0.856	2	2	34	502
bge	trec-covid	0.728	0.787	0.805	0.847	1	1	10	408
contriever	scifact	0.513	0.500	0.569	0.505	1	2	23	407
contriever	nfcorpus	0.296	0.327	0.373	0.404	2	6	25	417
contriever	fiqa	0.444	0.344	0.397	0.422	11	430	16	502
contriever	trec-covid	0.468	0.411	0.476	0.392	2	1	46	408

Table 8: HyDE geometry, computed from the cached generations. Cosines use unit vectors; ranks use the scoring of the actual run. "% rank better" is the share of queries whose first relevant document moves up when $f(q)$ is replaced by the HyDE vector.

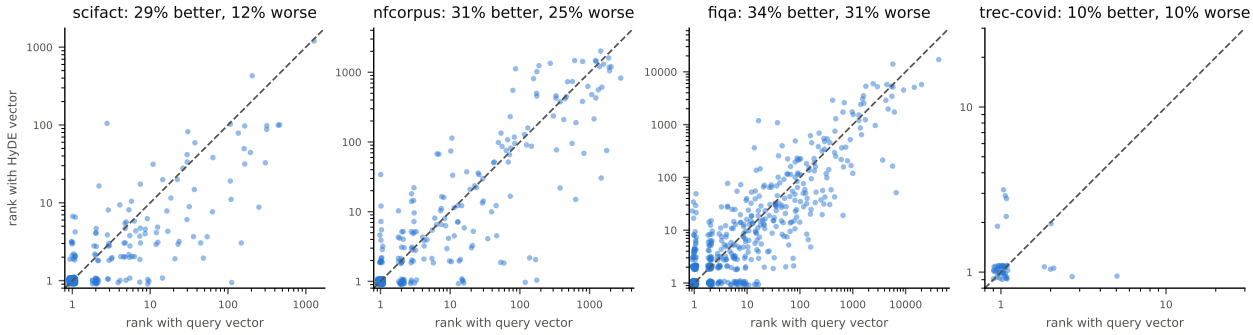


Figure 4: Rank of the first relevant document with the query vector (x) vs. with the HyDE vector (y), bge, one dot per query (log scales, slight jitter). Points below the diagonal are queries HyDE helps.

so the question itself is largely lost. On FiQA, where gpt-oss passages are longest (502 words) and often formatted as markdown tables, the passages are *farther* from the relevant documents than the query is (0.344 vs. 0.444), and the median rank of the first relevant document falls from 11 to 430. Cutting passages to 100 words reduces their norm to ≈ 1.65, which is why HyDE-Contriever-100w recovers FiQA. HyDE thus works when (i) the generator writes passages in the corpus's style and (ii) the encoder gives each vector an equal say. With bge both hold, and HyDE-bge is one of the best systems we tested.

6.3 Design choices for CCDB Search

The measurements above give a recipe, and a list of things that break HyDE:

- **Use a normalized encoder** (bge-style, unit vectors). With Contriever's unnormalized vectors the same passages made results *worse* than no HyDE at all.
- **Keep the query in the mean.** $f(q)$ anchors the vector to what was asked; when it was drowned out (Contriever) relevant documents fell from rank 11 to rank 430 on FiQA.
- **Make the prompt describe complaint narratives.** HyDE works because the passages look like corpus documents. The paper's prompts name the document type ("scientific paper passage", "financial article"); for the CCDB the prompt should ask for a consumer complaint

narrative, e.g. "Please write a consumer complaint to the CFPB describing this problem", in plain first-person prose of typical narrative length. gpt-oss's default style here (400–500-word markdown with headings and tables) is far from how consumers write, and on Contriever that mismatch cost most of the gain.

- **FiQA is the closest test, and the gain there was small.** FiQA (financial questions answered from forum posts) is the nearest of our collections to the CCDB. HyDE-bge improved on Dense there by only +0.007 nDCG@10, and 31 % of queries got worse. The financial domain alone does not guarantee a large benefit; matching the narrative style (previous point) is likely what matters.
- **Try fewer samples.** The eight passages were nearly the same vector (cosine 0.96), so 2–4 samples may give most of the benefit at lower cost. This should be tested on CCDB queries.
- **Plan for latency.** HyDE took 10–19 s per query here (the slowest of 8 parallel LLM calls), against about 10 ms for Dense. For an interactive search this needs a faster or smaller generator, streaming, caching of frequent queries, or a fallback that shows Dense results first.
- **Expect some queries to get worse.** 10–31 % of queries ranked their first relevant document lower with HyDE. Fusing HyDE with BM25 or Dense (as Hybrid does) is a cheap hedge.
- **Evaluate on the CCDB itself.** Gains ranged from +0.056 (SciFact) to −0.002 (TREC-COVID), so a small set of judged CCDB queries (e.g. labelled with the LLM judge of Section 5.1, after checking it against a few human labels) is needed before relying on it.

7 Cost

Table 7 gives the latencies. BM25 answers in 0.1–1.7 ms, Dense and Hybrid in 5–10 ms, SPLADE in 8–83 ms (sparse matrix product, no inverted index), and ColBERTv2/PLAID in 160–380 ms. The cross-encoder adds about 80 ms over the Hybrid top 50 and 150–250 ms over the top 100. The LLM methods are in another class: HyDE takes 10–19 s per query (the slowest of 8 parallel generations) and RankGPT 26–49 s (9 sequential windows). These were measured with 24 concurrent gateway requests, under which per-call latency rose about 1.6×, so they overstate single-user latency but not the order of magnitude. One-time setup also differs: on TREC-COVID, BM25 indexes in 7 s, bge-small encodes the corpus in 6 min, and ColBERTv2 needs 15 min to encode plus 75 min to build a 2.5 GB PLAID index. The LLM steps used 10,568 (HyDE), 11,022 (RankGPT) and 63,765 (judge) gateway calls, roughly 110M input tokens in total.

8 Threats to validity

- **Training overlap.** bge-small's fine-tuning mixture is not documented per dataset, so overlap with FiQA's or NFCorpus' training splits cannot be excluded. The ColBERTv2, SPLADE and Contriever results do not have this caveat.
- **Different LLMs.** gpt-oss-120b is not the model the HyDE and RankGPT papers used, and it is a much newer and stronger model than gpt-3.5. The RankGPT comparison shows what the method does with a current open-weight model, not a replication of the paper's numbers.
- **Lexical bias in the pools.** BEIR (Table 4) found TREC-COVID's pool still lexically biased despite its depth. J@10 shows our top-10 lists are mostly judged there (0.88 for both Dense and BM25), and filling the holes did not reverse any conclusion, but it remains a caveat.

- **Judge quality.** $\kappa = 0.53$ is moderate, and the judge disagrees with the NFCorpus labels. The augmented results are supporting evidence, not a replacement for the human qrels.
- **Multiple comparisons.** We ran 48 tests against BM25. With Bonferroni ($0.05/48 \approx 0.001$), only results marked ** survive; all conclusions above rest on those.
- **Small query set.** TREC-COVID has only 50 queries, although its main effects have $p < 10^{-4}$.
- **Fixed configurations and one machine.** Each system ran one configuration, chosen before seeing results; latency was measured on one laptop and is indicative only.

9 Verdict

- **SciFact:** yes, modestly for cheap methods (Hybrid +0.045, $p < 0.001$), substantially for LLM methods (HyDE-bge +0.090, RankGPT +0.086).
- **NFCorpus:** yes, modestly (Hybrid +0.042, RankGPT +0.064, both $p < 0.001$).
- **FiQA:** yes, strongly (Dense +0.150, RankGPT +0.176, $p < 0.001$); fusion with the weak BM25 list gives back a quarter of Dense's gain.
- **TREC-COVID:** yes, strongly (Hybrid +0.169, RankGPT +0.268, $p < 0.001$).

Semantic search helps most when queries are natural-language questions worded differently from the documents. A small supervised bi-encoder fused with BM25 delivers most of the gain for about 10 ms per query. LLM reranking delivers the best quality measured here, beyond the published gpt-4 numbers, at roughly 3,000–5,000 times the latency. HyDE's benefit depends on the generator and encoder working together, and does not carry over automatically to a new LLM.

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A Reproducibility

Run `python3 -m retrieval_eval.data.fetch_beir && .venv/bin/python run.py`. Each method is one module in `retrieval_eval/methods/` (e.g. `python -m retrieval_eval.methods.splade`); scoring, the LLM judge and the published reference numbers are in `retrieval_eval/evaluation/`. ColBERTv2 runs in a separate `.venv-colbert` (PyLate pins an older torch). The LLM steps need `databricks auth` access to the Unity Gateway; with the shipped `llm_cache/` they make no new calls (`-no-llm` skips them). Library versions, model revisions and probe output are in `RESULT.md`; judge labels are in `llm_labels/` and augmented qrels in `qrels_aug/`.